

SAARTHI: An AI-Powered Industrial Safety Copilot for Real-Time Excavator Hazard Detection

Sounak Chowdhury, Member 2, Member 3, Member 4, Member 5

Abstract—Write the abstract here.

Index Terms—Industrial Safety, Computer Vision, YOLOv8, Excavator, Hazard Detection, Edge AI, Large Language Models

I. INTRODUCTION

Write the introduction here.

II. RELATED WORK

Securing heavy-machinery operations sits at the intersection of computer vision, physiological monitoring, and adaptive decision-making. Broader surveys of industrial safety still tend to focus on rule-based alarms and single-sensor approaches, and even where hybrid deep learning architectures have made real progress on driver drowsiness detection in automotive settings, they are built around a single modality. That assumption breaks down on a construction site, a mine, or a factory floor, where operators differ widely in experience and ability and hazards rarely show up one at a time. This section works through that prior literature and situates the design choices behind SAARTHI against it.

A. Threshold-Based and Infrastructure-Centric Safety Systems

Most collision-prevention systems on active worksites still come down to proximity alarms and geofencing: fixed ultrasonic or radar sensors mounted on the machine, scanning continuously and comparing whatever they detect against a static, manually set threshold, with no regard for who's actually operating the machine. In practice, those fixed thresholds tend to misfire constantly on busy sites, where workers and equipment are always moving through the sensor's range. Worse, the same threshold gets applied to every operator regardless of experience or physical ability, so an experienced operator ends up tuning out the alarms while a newer one may not get warned early enough.

Newer cabin-integrated systems have brought computer vision into the mix, using object detection to pick out workers, vehicles, and structural hazards around the machine, and drawing on signals like facial landmark drift, eyelid closure, or chassis tilt to flag unsafe states. But most of these still work in isolation. A camera watching for fatigue has no idea there's a person standing in the machine's blind spot, and a proximity sensor has no way of knowing the operator is falling asleep. Because these hazards arise from genuinely different physical processes, no single detector is well positioned to catch all of them—which is really the argument for fusing

several lightweight signals rather than trying to build one model that does everything.

Conventional alarm-based systems, built for fairly controlled, homogeneous environments, don't hold up well once the operator population and the worksite itself become this varied.

- **Rigidity and operator homogeneity.** A fixed-threshold system treats every operator identically, regardless of experience, sensory ability, or fatigue tolerance. In practice that means the same threshold that produces alarm fatigue in a veteran operator will under-warn someone newer to the job—there's no way to individualize the safeguard without changing the underlying logic entirely.
- **Deployment complexity and accessibility gaps.** Combining vision, physiological, and environmental sensing into one cabin system is already a nontrivial integration problem, and most existing architectures weren't designed with that convergence in mind. Accessibility tends to be an afterthought too: operators with hearing or visual impairments are rarely considered when alert delivery is designed almost exclusively around audio cues.

B. Fatigue-Centric and Vision-Based Defenses

A separate line of work moves detection logic onto the operator's own physiological state rather than the surrounding environment. Here the typical approach is a camera-based agent watching the driver's face—tracking eye-aspect-ratio, yawning frequency, head pose—to infer drowsiness before it becomes dangerous.

These systems are reasonably good at what they do, but that's also their limitation: they know almost nothing about what's happening outside the cabin. A fatigue detector can't tell you there's a worker nearby, or that the terrain has become unstable, or that the machine itself is tilting toward a rollover. It's built to catch one kind of hazard and stays blind to the rest.

- **Modality opacity.** Vision-only systems generally ignore machine-state telemetry altogether, even though rollover risk from chassis inclination is well documented in the tilt-sensing literature. A fatigue model has no access to that data by design, so it simply can't detect a compound hazard—drowsiness plus instability, say—even when both are present at once.
- **The context gap.** These tools are built around a single-hazard focus and stay that way in deployment. They fire only on drowsiness cues, missing nearby obstacles or

unstable ground entirely. By the time a purely fatigue-driven alert goes off, the operator may already be in a situation involving more than just tiredness.

C. Machine Learning in Industrial Safety

As static, rule-based systems have repeatedly failed to keep up with a workforce this varied in experience and accessibility needs, machine learning has become the default approach for adaptive intervention.

1) *Deep Learning and Object Detection Models*: Deep learning dominates recent work in this space. YOLO-based detectors, for instance, perform well in real time for identifying blind-spot obstacles and nearby workers. On the physiological side, convolutional facial-landmark models track eye and head movement continuously to catch fatigue before it becomes critical, and gradient-boosted models like XGBoost have proven useful for context-aware risk scoring, weighing environmental and operational variables against historical incident records.

The catch is that these models are almost always deployed as separate pipelines. Nobody is fusing their outputs into one coherent decision—which leaves a human supervisor to piece together several unrelated alerts in real time.

- **Fragmented decision-making.** Running fatigue, obstacle, and tilt detection as three independent systems means a supervisor has to mentally combine uncorrelated alerts on the fly. That adds real cognitive load, slows reaction time, and raises the odds that a compound hazard slips through unnoticed.
- **Static intervention policy.** Even where detection is sophisticated, the response usually isn't: cabins tend to issue the same alert regardless of context, which is exactly the kind of rigid logic an adaptive, constraint-respecting policy is meant to replace.

2) *Lightweight Rule-Based Models*: Some practitioners have gone the other direction, relying on fixed-weight scoring rules or manually curated alert matrices instead. These are cheap to implement and integrate easily into legacy cabin electronics, which is a real advantage.

The tradeoff is that manual thresholds don't adapt well to individual variation—different operator experience levels, different accessibility needs, changing site conditions. Keeping thresholds current across a large fleet turns into an ongoing manual calibration problem, which doesn't scale. Reinforcement learning offers a way around this, letting a system learn its own intervention policy while staying inside predefined safety bounds, though getting RL to behave safely and explainably in a live industrial setting is still an open problem—earlier work has shown that unconstrained policy learners can behave unpredictably during exploration, especially when safety constraints aren't built into training from the start.

D. The Gap Analysis

Taken together, the literature points to a fairly clear gap: nobody has really combined multiple hazard-detection modalities, adaptive decision-making, explainability, and operator-specific accessibility into one system. Most work picks one

of these and pushes it further, while the others—particularly personalization—stay largely unaddressed.

SAARTHI is built around closing that specific gap. It fuses fatigue, blind-spot, tilt, and contextual risk signals into a single Reinforcement Learning policy operating under safety constraints, giving it the kind of adaptive intervention selection that has mostly been left to human judgment after the fact, while keeping the decision process explainable enough for safety-critical use. Rather than treating accessibility as something bolted on afterward, operator-specific delivery preferences are built directly into the Hybrid Decision Engine—so the same underlying risk score can reach an operator through haptic, visual, or audio channels depending on what they actually need, without changing the safety guarantees behind the decision itself.

III. SYSTEM MODEL AND THEORETICAL FRAMEWORK

To ground the design in real operating conditions, we first characterize the driver and the machine's surrounding environment as they evolve in real time. On-device deployment inside a moving vehicle comes with hard limits—a tight latency budget and modest compute—and these constraints shape everything that follows. They rule out streaming raw, uncompressed HD frames, and they rule out running heavy spatio-temporal deep networks directly at the edge, which is exactly where traditional vehicle-tracking pipelines fall short.

SAARTHI takes a different route. It represents the operator and machine state through localized vision primitives paired with synchronized telemetry. This approach compresses the high-frequency video and sensor streams into a compact risk feature vector while preserving the information that actually matters for hazard detection: the signatures of micro-sleep, blind-spot intrusions, and axis tilt. The sections below lay out the hazard threat model, the reasoning behind real-time risk estimation, and the localized telemetry features used to flag imminent operational anomalies.

TABLE I
COMPARATIVE ANALYSIS OF OPERATOR SAFETY PARADIGMS

Methodology	Hardware Class	Latency	Primary Limitation
Cloud-Based Telematics	Server + Cellular Node	High	Connectivity reliance in remote sites (e.g. mines)
Wearable Biosensors	Smartwatch / EEG Cap	Low	Operator discomfort, hygiene, compliance drop-off
Heavy Deep Learning (CNN/ViT)	Edge (Jetson/TPU)	Medium	High computational complexity, thermal output
Basic Machine Thresholds	Machine ECU	Low	Rigidly generic, no behavioral insight
SAARTHI (Proposed)	Commodity CPU	Low	Requires 30 s initial operator calibration

A. The Hazard Model

Hazards in industrial vehicle environments fall into two broad classes, separated by how visible they are and how much computational reasoning reliable detection demands.

1) *Type 1: Overt Operational Hazards*: A Type 1 hazard occurs whenever a physical safety boundary or operational constraint is breached—a pedestrian stepping into a blind spot, an operator letting go of the wheel with both hands, or any incursion into a predefined safety zone. Like conventional rule-based anomaly detection, these events carry clear, observable visual signatures, which makes them deterministic and comparatively straightforward to catch.

SAARTHI detects them in real time by checking the bounding boxes produced by the YOLO detector against predefined restricted regions. The moment an object crosses into a prohibited zone, the system flags a Type 1 hazard and triggers the appropriate safety response.

2) *Type 2: Covert Micro-behavioral Hazards*: Type 2 hazards are subtle behavioural or contextual anomalies, and they are far harder to spot. Because they tend to blend into normal operation, simple spatial rules or threshold-based vision cannot isolate them. An operator in the early stages of fatigue, for instance, may sit with an apparently normal posture and open eyes while slowly drifting into reduced alertness. Machine vibration or tilt behaves the same way: each reading may stay within acceptable bounds on its own, yet together the readings point to an abnormal operating state.

Since these situations look almost identical to a safe state on the surface, vision alone is not enough. SAARTHI instead examines the statistical character and temporal evolution of the operator’s micro-movements—blink-rate variation, gaze drift, head pose—alongside machine telemetry over time. This deeper behavioural view surfaces latent hazards before they escalate into critical incidents.

B. Anomaly Theory in the Physiological and Kinematic Sense

We monitor the operator’s state in real time using facial-landmark coordinate features. The core primitive is the Eye Aspect Ratio (EAR), a per-frame measure of how open the eyes are:

$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2||p_1 - p_4||} \quad (1)$$

Here p_i denotes the 2D position of an eye landmark.

For an alert operator, the EAR sequence stays fairly stable around an open-eye mean μ_{alert} , and the fluctuations that do occur—ordinary physiological blinks, small head-pose shifts, minor lighting changes—sit within a natural range. The resulting eye-state distribution is therefore roughly bell-shaped and symmetric.

The framework rests on a simple premise: the onset of fatigue introduces specific, non-Gaussian distortions into this physiological stream.

- **Microsleep artifacts.** Prolonged eye closures—the hallmark of severe drowsiness—extend well beyond the normal blink window. They appear as rapid collapses in the EAR stream, the phenomenon the PERCLOS metric is designed to quantify, and they look nothing like the smooth, brief dip of an ordinary blink.
- **Heavy-tailed distributions.** These long closures skew the eye-state distribution and thicken its lower tail. As the operator jerks awake and tries to recover the open-eye

mean μ , the distribution no longer looks symmetric—its heavier tail is the statistical fingerprint of fatigue.

Kinematic anomalies—rollover risk, blind-spot intrusion—are handled on a separate track, driven by machine telemetry and spatial bounding boxes. When the machine is moving ($v \neq 0$) and the tilt angle θ exceeds a critical set-point θ_{crit} , the response is deterministic rather than behavioural: a Tier-1 safety intervention fires immediately, independent of the standard behavioural policies.

C. Mathematical Formalization of Features

To turn these physiological cues into something a classifier can act on, the system extracts a feature vector $\mathbf{x}$ from a sliding temporal window $W = \{e_1, e_2, \dots, e_w\}$ of raw EAR samples. Table II lists the notation used throughout the derivation.

TABLE II
MATHEMATICAL NOTATIONS AND DEFINITIONS. *These symbols define the statistical moments used to construct the feature vector $\mathbf{x}$, mapping raw physiological data into the decision space.*

Symbol	Definition
W	Sliding temporal window of size w
e_i	Individual EAR sample at frame i
μ	Mean EAR (first moment)
σ	Standard deviation (second moment)
S_k	Skewness (third standardized moment)
K_u	Excess kurtosis (fourth standardized moment)
E_{roc}	EAR rate of change
$\Phi(\mathbf{x})$	Non-linear mapping to the XGBoost feature space

1) *Skewness (S_k)*: Skewness measures the asymmetry of the signal’s distribution. For an alert operator the eye-openness distribution stays close to symmetric over time ($S_k \approx 0$). As fatigue sets in, micro-sleeps and long closures pull the EAR values toward the low, closed-eye end, producing an asymmetric lower tail.

$$S_k = \frac{\frac{1}{w} \sum_{i=1}^w (e_i - \mu)^3}{\left(\frac{1}{w} \sum_{i=1}^w (e_i - \mu)^2\right)^{3/2}} \quad (2)$$

2) *Kurtosis (K_u)*: Kurtosis is the framework’s primary discriminator for deep drowsiness. It measures the tailedness of the distribution:

$$K_u = \frac{\frac{1}{w} \sum_{i=1}^w (e_i - \mu)^4}{\left(\frac{1}{w} \sum_{i=1}^w (e_i - \mu)^2\right)^2} - 3 \quad (3)$$

Subtracting 3 yields excess kurtosis, normalized against a standard normal distribution. Values far from 0 flag the kind of outliers consistent with microsleep episodes rather than ordinary blinking.

3) *EAR Rate of Change (E_{roc})*: The EAR rate of change captures the fast transitions typical of a fatigue episode—a sudden jerk awake, a rapid head drop:

$$|E_{roc}| = |e_t - e_{t-k}| \quad (4)$$

Here $k = 5$ sets the stride, measuring displacement across several frames. A large $|E_{roc}|$ signals a rapid eye snap or an abrupt physiological shift—motion you rarely see in calm, alert operation, where the state drifts slowly.

TABLE III

SENSITIVITY ANALYSIS OF THE FEATURES. *Each feature maps to a particular type of anomaly. Mean EAR supplies physiological background, while kurtosis and skewness pick up the non-Gaussian behaviour that characterizes the onset of fatigue.*

Feature	Physiological Meaning	Detection Function
Mean EAR (μ)	Baseline openness	Establishes the EAR baseline, providing physiological context.
Std. dev. (σ)	Blink volatility	Stability check: high variance flags erratic blinking or an unstable head pose.
Kurtosis (K_u)	Drowsiness burstiness	Primary discriminator: detects deep micro-sleeps; longer closures raise kurtosis.
Skewness (S_k)	Distribution symmetry	Fatigue artifacts: detects the closed-eye asymmetry in the distribution.
Rate of change (E_{roc})	Physiological velocity	Flags rapid eye movements or sudden head drops.

D. Defining “Real-Time” for Edge AI Safety

One of the first and crucial steps in Edge AI Safety is to clearly define what is meant by “Real-Time”. However, the statistical physiological analysis requires a minimum number of w frames to be able to extract a meaningful distribution to detect fatigue.

This paper defines Real-Time in the context of Operator Safety as:

$$T_{\text{inference}} \ll T_{\text{feature_window}} < T_{\text{hazard_onset}} \quad (5)$$

The data collection time ($T_{\text{feature_window}}$) is not the critical metric, but rather the Inference Latency ($T_{\text{inference}}$) is the critical metric for Real-Time in the context of Operator Safety. The continuous feature extraction window is implemented with a moving buffer. When a Critical Fatigue event occurs, or when an operator shows signs of Critical Fatigue, a verdict should be available from the Hybrid Decision Engine ($T_{\text{hazard_onset}}$). With a considerably lower algorithmic processing time for the inference (milliseconds) compared to the time it takes the human to react in order to prevent an accident (T_{human}), the system can issue a customised warning earlier. To avoid this decision logic becoming the performance obstacle, SAARTHI will reduce $T_{\text{inference}}$ by using lightweight models optimized for the edge.

IV. CONCLUSION AND FUTURE WORK

This paper introduced SAARTHI, a personalized AI safety copilot for heavy-machinery operators facing multi-hazard industrial risks. Moving away from the siloed, rule-based alarms that dominate the field, the system fuses hazard detection into a single Hybrid Decision Engine governed by a constrained Reinforcement Learning policy—an approach that tries to reconcile consistent safety enforcement with the fact that no two operators have quite the same needs.

A. Summary of Contributions

Three things stand out from the design and integration of the system.

- **Multi-modal risk fusion.** Combining fatigue, blind-spot, tilt, and contextual risk signals into one real-time safety score captures compound hazards that any single detector would miss on its own. The individual detectors supply the sensory groundwork, but it’s the fusion logic in the Hybrid Decision Engine that actually decides when intervention is warranted.
- **Balancing safety and personalization.** The PPO-based policy weighs predefined safety constraints against operator-specific delivery preferences, choosing interventions based on accessibility needs, experience level, and stated alert preferences. That’s a meaningfully different approach from the rigid, uniform alarm thresholds most systems still rely on, and it shows that transparent, explainable safety copilots are achievable without giving up consistency.
- **Scalable and accessible deployment.** Centralizing personalization in operator profiles, and surfacing decisions through a supervisor dashboard, gives SAARTHI an edge over fragmented, single-purpose safety hardware—it scales across construction, mining, and manufacturing fleets while closing an accessibility gap those single-purpose devices never addressed. The Explainable AI layer on top of that keeps both operators and supervisors able to trust why any given intervention happened.

B. Limitations and Future Work

The framework is a solid starting point, but the design process surfaced a few areas that clearly need more work.

- **Automatic operator identification.** Right now the system assumes an operator manually selects their profile at shift start—accessibility settings, preferred language, alert preferences, all of it. On a busy or rotating crew, that step gets skipped often enough that the cabin defaults to generic settings that don’t actually match whoever’s in the seat. Facial recognition for automatic operator identification is the obvious fix: detect who’s in the cabin, load their profile—including whatever language settings they’ve already configured—without requiring a manual login. It closes the gap between what the system is meant to do and what actually happens in practice.
- **Wearable sensor fusion for fatigue robustness.** The fatigue model currently leans almost entirely on facial-landmark tracking, which struggles under poor cabin lighting, camera occlusion, or when an operator is wearing a dust mask or safety goggles. Bringing in wearable physiological sensors—heart-rate variability bands, EEG headsets—to fuse with the existing vision-based estimate should make the fatigue score noticeably more robust to exactly those failure modes.
- **Field validation and policy hardening.** So far the reinforcement learning policy has only been trained and tested in simulated, controlled cabin conditions. Real deployment across construction, mining, and manufacturing sites will expose it to dust, vibration, and lighting variability that simulation just doesn’t capture well, and that gap needs to be closed before the policy can be

trusted at scale. The plan is a supervised Field Calibration phase—collecting real-world incident and near-miss data to refine the XGBoost context model’s risk thresholds and retrain the PPO policy under actual operating conditions, while keeping the predefined safety constraints intact throughout.